

Author Contributions Statement

Adaptive Forgetting Factor for Drifting Inverter-Based Resource Penetration in Model-Based Protection Systems

Author contributions follow the CRediT taxonomy. **AVE** = Arun V. Eluvathingal; **RKP** = R. K. Prasad (supervisor).

CRediT role	AVE	RKP
Conceptualisation	Lead	Supporting
Methodology	Lead	Supporting
Software	Lead	—
Validation	Lead	Supporting
Formal analysis	Lead	Supporting
Investigation	Lead	—
Resources	—	Lead
Writing – original draft	Lead	—
Writing – review & editing	Lead	Lead
Supervision	—	Lead

Detailed statement.

Arun V. Eluvathingal (AVE): Identified the stationarity assumption gap in existing SAMBPS implementations. Derived the steady-state tracking-bias formula and the tracking–noise trade-off bound. Designed the CUSUM detector and logistic λ -adaptation law; calibrated parameters over solar diurnal, cloud transient, wind, and topology-step archetypes. Proved Proposition 1 (conditioning bound). Implemented the Monte Carlo framework (5 000 trials per scenario, 20 000 total) and analysed all results. Wrote all sections of the manuscript.

R. K. Prasad (RKP): Provided research direction and ANDES/RTDS simulation infrastructure. Reviewed the CUSUM design and stability proof. Edited and approved the final manuscript.

Plain-text portal version:

AVE: Conceptualisation (Lead); Methodology (Lead); Software (Lead); Validation (Lead); Formal analysis (Lead); Investigation (Lead); Writing – original draft (Lead); Writing – review & editing (Lead).

RKP: Conceptualisation (Supporting); Methodology (Supporting); Validation (Supporting); Formal analysis (Supporting); Resources (Lead); Writing – review & editing (Lead); Supervision (Lead).